BeeQ: retrospective MLP class-weight sensitivity

Reproducibility and editorial supplement

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Status. Additional analyses A/B retain the original unweighted baseline; external evaluations are retrospective. The reviewed main manuscript source is unavailable in this checkout.

1 Methods

The original MLP baseline used StandardScaler followed by scikit-learn 1.9.0 MLPClassifier, with ReLU activation, Adam, max_iter=2000, early_stopping=True, validation_fraction=0.1, and random_state=20260824. It was trained without class balancing. The implementation omitted sample weights when the MLP was introduced; no contemporaneous rationale establishes why. The recorded library supported sample_weight, so a library restriction cannot be cited as the explanation. LR, RF and the three SVC-based baselines retained their original weighting mechanisms. This initial asymmetry remains a limitation of the original comparison.

We conducted an additional retrospective sensitivity analysis using the same X10 features, molecules and frozen partitions. For every training fit, class weights were calculated as w_c=N_fit/(2 n_c,fit) from that fit's labels and passed only to the MLP as sample weights. The scaler and all evaluation metrics remained unweighted; no resampling was used. Analysis A fixed each outer fold's hyperparameters and the final configuration to those selected by the original unweighted MLP. Analysis B repeated the full 12-candidate search over hidden_layer_sizes={(32,), (32,16)}, alpha={0.0001,0.001,0.01}, and learning_rate_init={0.0003,0.001}, using five frozen outer folds and four inner cluster-disjoint StratifiedGroupKFold partitions with seeds 20260824+outer_fold. Mean unweighted inner AUROC selected the first highest-scoring ParameterGrid candidate.

For each variant and outer training partition, thresholds were estimated separately from complete inner OOF predictions by maximizing MCC, then balanced accuracy, then minimizing the absolute threshold and finally the threshold itself. Final configuration selection used the five frozen Development folds; its OOF predictions determined a final threshold under the same rule, followed by training on all Development. Selection OOF metrics are not an additional unbiased estimate of generalization. All configurations, thresholds and model artifacts were frozen before the new input CSV evaluations of the historical holdout and CR8. These datasets and historical predictions had already been observed, so the extension is retrospective, not a new blind evaluation.

Native early stopping was preserved in both variants. Its stratified 10% molecule reservation is not cluster disjoint, and StandardScaler is fitted before that reservation. Weights are derived from the enclosing CV training fit, including this native stopping reservation; no CV evaluation, holdout or CR8 labels contribute to the class counts. The native implementation carries these weights into stopping accuracy and normalizes both the loss gradient and L2 penalty by each minibatch's weight sum. Therefore Analysis A measures the effect of native weighted training, including its stopping behavior, rather than a pure loss-gradient effect with an invariant stopping trajectory. Architecture, optimizer, alpha, batch size, tolerance and stopping settings were not changed.

Differences were paired on the same outer folds. For comparison with the original project statistics, we also used 2,000 paired Butina-cluster bootstrap resamples of fixed pooled OOF scores (seed 20260824). These descriptive intervals condition on the fitted predictions, do not repeat training or selection, and do not constitute multiplicity-adjusted hypothesis tests. Cross-fold score calibration and overlapping training sets limit interpretation of pooled statistics and five-fold consistency.

$$w_c = \frac{N_{\text{fit}}}{2n_{c,\text{fit}}}, \qquad w_i = w_{y_i}, \quad c \in \{0, 1\}.$$

2 Results

The original MLP was reproduced before interpreting balancing: all five nested OOF folds, selected configurations, thresholds, historical-holdout predictions and CR8 predictions agreed with the recorded campaign (maximum absolute score difference 1.89e-15; tolerance 1e-10). Its Development outer-fold AUROC was 0.677 ± 0.124 and AUPRC 0.596 ± 0.138; historical-holdout AUROC was 0.462. Thus the reviewed approximate values refer to the outer-fold mean and holdout, respectively; pooled nested OOF AUROC was 0.668.

With original hyperparameters fixed (Analysis A), outer-fold AUROC became 0.688 ± 0.100 and AUPRC 0.609 ± 0.119. With full retuning (Analysis B), AUROC was 0.679 ± 0.100 and AUPRC 0.594 ± 0.112. Paired mean AUROC changes were +0.0113 and +0.0021, respectively. At the fold-specific MCC thresholds, pooled toxic recall changed from 75/222 to 79/222 and 75/222, while false positives increased from 15/490 to 39/490 and 36/490. Recall improved in four folds but fell from 12/37 to 1/37 in the third fold in both balanced variants. Consequently sensitivity did not improve consistently. Pooled MCC decreased from 0.428 to 0.344 and 0.338.

The descriptive pooled AUROC bootstrap intervals included zero for both balanced-minus-original comparisons (A: -0.0242 to 0.0356; B: -0.0369 to 0.0258). For retuned pooled AUPRC the difference was -0.0474, with a descriptive interval of -0.0989 to -0.0009. This negative pooled result differs from the small outer-fold mean change and must not be described as a global discrimination improvement. No causal explanation of the MLP's overall inferiority is established by these results.

On the historical holdout, AUROC/AUPRC changed from 0.462/0.302 to 0.481/0.332 in A and 0.482/0.329 in B. Using separately frozen thresholds (original 0.524583, A 0.536224, B 0.528334), toxic recovery increased from 4/49 to 11/49 and 13/49, while specificity fell from 128/132 (0.970) to 116/132 (0.879) and 110/132 (0.833). False positives rose from 4 to 16 and 22. Discrimination therefore remained weak, whereas the major practical change was increased positive decisions with a specificity cost.

In CR8, both balanced variants detected Isocycloseram, which the original missed; all variants missed Ethiprole. Fluoxapiprolin became an additional false positive, while Florylpicoxamid remained a false positive. AUROC/AUPRC decreased from 0.833/0.583 to 0.667/0.450 in A and 0.583/0.367 in B. Recovering one of two toxic compounds cannot establish population sensitivity or robust regional generalization.

All 693 fits stopped before 2,000 iterations (12-77 iterations) without recorded training warnings. This verifies finite training and the configured stopping behavior, not optimization to a global or stationary solution. Native stopping reservations shared 14-32 clusters with their gradient-training subsets. An independent complete rerun reproduced all tabular outputs exactly. The balanced retuned MLP remains numerically below every other BeeQ model in mean outer AUROC; RF remains highest at 0.748. The quantum-versus-matched-RBF results are unaffected. The findings therefore preserve BeeQ's central comparative conclusions while qualifying the original MLP limitation: balancing improves retrospective toxic recovery at the price of more false positives and does not resolve its weak discrimination.

3 Development summary

Table S1a: AUROC/AUPRC: outer-fold mean and sample SD. BA/MCC: pooled nested OOF decisions. A fixes original hyperparameters; B repeats the search.

Variant	AUROC mean/SD	AUPRC mean/SD	Pooled BA	Pooled MCC
Original	0.6771 ± 0.1245	0.5961 ± 0.1383	0.654	0.428
Balanced A	0.6884 ± 0.1005	0.6086 ± 0.1186	0.638	0.344
Balanced B	0.6792 ± 0.0997	0.5942 ± 0.1119	0.632	0.338

Table S1b: Pooled nested Development decisions (490 negative, 222 toxic).

Variant	Sens.	Spec.	TP	FN	TN	FP
Original	0.338	0.969	75	147	475	15
Balanced A	0.356	0.920	79	143	451	39
Balanced B	0.338	0.927	75	147	454	36

4 External evaluation and frozen thresholds

Table S2a: historical_holdout: discrimination and frozen-threshold metrics.

Variant	AUROC	AUPRC	BALANCED_ACCURACY	MCC	THRESHOLD
Original	0.462	0.302	0.526	0.111	0.524583
Balanced A	0.481	0.332	0.552	0.129	0.536224
Balanced B	0.482	0.329	0.549	0.111	0.528334

Table S2b: historical_holdout: toxic recovery and specificity.

Variant	SENSITIVITY	SPECIFICITY	TP	FN	TN	FP
Original	0.082	0.970	4	45	128	4
Balanced A	0.224	0.879	11	38	116	16
Balanced B	0.265	0.833	13	36	110	22

Table S2c: CR8: discrimination and frozen-threshold metrics.

Variant	AUROC	AUPRC	BALANCED_ACCURACY	MCC	THRESHOLD
Original	0.833	0.583	0.417	-0.218	0.524583
Balanced A	0.667	0.450	0.583	0.149	0.536224
Balanced B	0.583	0.367	0.583	0.149	0.528334

Table S2d: CR8: toxic recovery and specificity.

Variant	SENSITIVITY	SPECIFICITY	TP	FN	TN	FP
Original	0.000	0.833	0	2	5	1
Balanced A	0.500	0.667	1	1	4	2
Balanced B	0.500	0.667	1	1	4	2

5 Outer folds and selected hyperparameters

Table S3a: Original: outer-fold metrics.

Fold	AUROC	AUPRC	BA	MCC	Sens.	Spec.	THRESHOLD
1	0.489	0.405	0.585	0.283	0.200	0.969	0.524038
2	0.637	0.521	0.625	0.325	0.324	0.926	0.539855
3	0.792	0.659	0.658	0.482	0.324	0.991	0.520559
4	0.683	0.628	0.619	0.416	0.237	1.000	0.527747
5	0.785	0.768	0.764	0.582	0.592	0.935	0.513438

Table S3b: Original: outer-fold class counts and decisions.

Fold	NEGATIVE	POSITIVE	TP	FN	TN	FP
1	97	40	8	32	94	3
2	68	37	12	25	63	5
3	110	37	12	25	109	1
4	122	59	14	45	122	0
5	93	49	29	20	87	6

Table S3c: Balanced A: outer-fold metrics.

Fold	AUROC	AUPRC	BA	MCC	Sens.	Spec.	THRESHOLD
1	0.536	0.450	0.552	0.113	0.300	0.804	0.561234
2	0.645	0.532	0.639	0.352	0.351	0.926	0.636153
3	0.781	0.631	0.514	0.143	0.027	1.000	0.650128
4	0.713	0.680	0.662	0.420	0.373	0.951	0.555593
5	0.767	0.749	0.768	0.566	0.633	0.903	0.541755

Table S3d: Balanced A: outer-fold class counts and decisions.

Fold	NEGATIVE	POSITIVE	TP	FN	TN	FP
1	97	40	12	28	78	19
2	68	37	13	24	63	5
3	110	37	1	36	110	0
4	122	59	22	37	116	6
5	93	49	31	18	84	9

Table S3e: Balanced B: outer-fold metrics.

Fold	AUROC	AUPRC	BA	MCC	Sens.	Spec.	THRESHOLD
1	0.536	0.450	0.552	0.113	0.300	0.804	0.561234
2	0.645	0.532	0.639	0.352	0.351	0.926	0.636153
3	0.781	0.631	0.514	0.143	0.027	1.000	0.650297
4	0.668	0.608	0.640	0.411	0.305	0.975	0.560083
5	0.767	0.749	0.768	0.566	0.633	0.903	0.541755

Table S3f: Balanced B: outer-fold class counts and decisions.

Fold	NEGATIVE	POSITIVE	TP	FN	TN	FP
1	97	40	12	28	78	19
2	68	37	13	24	63	5
3	110	37	1	36	110	0
4	122	59	18	41	119	3
5	93	49	31	18	84	9

Table S4: Selected outer-fold configurations and thresholds.

Variant	Fold	Layers	Alpha	Rate	THRESHOLD
Original	1	[32, 16]	0.0100	0.0010	0.524038
Original	2	[32, 16]	0.0100	0.0010	0.539855
Original	3	[32, 16]	0.0010	0.0010	0.520559
Original	4	[32, 16]	0.0100	0.0010	0.527747
Original	5	[32, 16]	0.0001	0.0010	0.513438
Balanced A	1	[32, 16]	0.0100	0.0010	0.561234
Balanced A	2	[32, 16]	0.0100	0.0010	0.636153
Balanced A	3	[32, 16]	0.0010	0.0010	0.650128
Balanced A	4	[32, 16]	0.0100	0.0010	0.555593
Balanced A	5	[32, 16]	0.0001	0.0010	0.541755
Balanced B	1	[32, 16]	0.0100	0.0010	0.561234
Balanced B	2	[32, 16]	0.0100	0.0010	0.636153
Balanced B	3	[32, 16]	0.0100	0.0010	0.650297
Balanced B	4	[32, 16]	0.0001	0.0003	0.560083
Balanced B	5	[32, 16]	0.0001	0.0010	0.541755

Original and A final configuration: layers (32,16), $\alpha = 0.01$, initial learning rate 0.0003. B final configuration: layers (32,16), $\alpha = 0.001$, initial learning rate 0.001. Final selection uses the five frozen Development folds, not external labels.

6 CR8 compound predictions

Table S5a: CR8 scores (positive class = toxic).

Compound	Truth	Original score	Balanced A score	Balanced B score
Isotianil	0	0.428	0.477	0.480
Fluoxapiprolin	0	0.487	0.573	0.575
Ethiprole	1	0.489	0.519	0.520
Cyclobutrifluram	0	0.471	0.520	0.522
Isoprothiolane	0	0.373	0.451	0.454
Isocycloseram	1	0.511	0.576	0.573
Spiropidion	0	0.415	0.466	0.468
Florylpicoxamid	0	0.535	0.645	0.653

Table S5b: CR8 decisions at separately frozen thresholds; only two compounds are toxic.

Compound	Truth	Original pred	Balanced A pred	Balanced B pred
Isotianil	0	0	0	0
Fluoxapiprolin	0	0	1	1
Ethiprole	1	0	0	0
Cyclobutrifluram	0	0	0	0
Isoprothiolane	0	0	0	0
Isocycloseram	1	0	1	1
Spiropidion	0	0	0	0
Florylpicoxamid	0	1	1	1

7 Paired cluster bootstrap

Table S6: 2,000 paired cluster resamples of pooled OOF predictions. Intervals are descriptive; the last column is not a p-value.

Comparison	METRIC	Delta	CI low	CI high	P_DELTA_GT_0
Balanced A - Original	AUROC	0.005	-0.024	0.036	0.621
Balanced A - Original	AUPRC	-0.026	-0.073	0.018	0.122
Balanced B - Original	AUROC	-0.006	-0.037	0.026	0.351
Balanced B - Original	AUPRC	-0.047	-0.099	-0.001	0.022
RF - Balanced B	AUROC	0.095	0.024	0.171	0.997
RF - Balanced B	AUPRC	0.146	0.073	0.213	1.000

8 Training audit and reproducibility

Table S7: All fits stopped before max_iter=2000; no recorded training warnings. This is not proof of global optimization.

Variant	FITS	Min iter	Max iter	WARNINGS
Balanced A	31	14	52	0
Balanced B	331	12	77	0
Original	331	12	62	0

Python 3.12.8; scikit-learn 1.9.0; NumPy 2.5.1; pandas 2.3.3. Model seed 20260824; inner seeds 20260825–20260829; bootstrap seed 20260824. Full-Development class weights: 0.7265306122 (negative), 1.6036036036 (toxic). All five outer and twenty inner CV splits are cluster disjoint. Native early-stopping reservations are not: 14–32 clusters overlap their gradient subsets. No resampling or evaluation weighting was used. All CSV outputs match an independent complete rerun exactly.

Repeat from the repository root with its historical environment:

```
.\.venv\Scripts\python.exe -m src.mlp_sensitivity
# Add --output-dir NEW_PATH; existing bundles are never overwritten.
.\.venv\Scripts\python.exe -m src.mlp_sensitivity_report
# Add --run NEW_PATH --output-dir docs/mlp_sensitivity
```

Authoritative bundle: `results/mlp_sensitivity/full_20261009_v2`.

The bundle contains input/source hashes, exact split memberships, candidate-level AUROCs, selected parameters, threshold OOF predictions, full fit/stopping histories, model artifacts, post-freeze external predictions and metrics, paired statistics, a sensitivity figure and integrity manifests. The separate manuscript replacement file maps this additional analysis to Methods III-D, Results IV-A/IV-D and the original baseline table without deleting the initial limitation.

Library references: Official MLPClassifier API documentation and the locally hashed 1.9.0 implementation. No runtime dependency was upgraded.